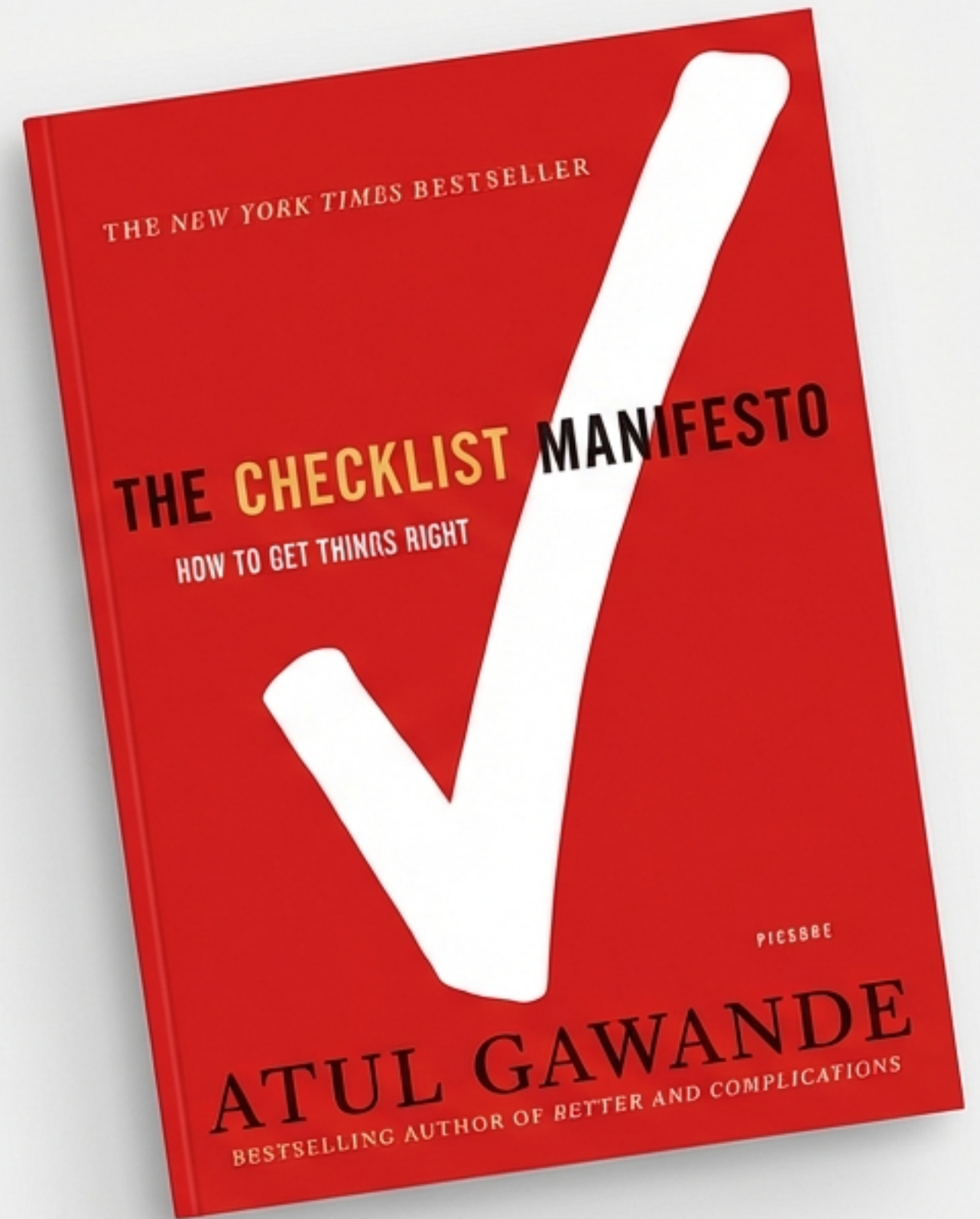


The Checklist Manifesto

How to Get Things Right
in a Complex World

Based on the work of Atul Gawande.



The solution to extreme complexity is
not more expertise, but a humble tool.

Stable ✓

Crash ✓

Airway established ✓

Fluids pushed ✓

Rapid Transport ✓



The Missed Question:
What was the weapon?
A Bayonet.

**“Your mind doesn’t
think of a bayonet
in San Francisco.”**



Why We Fail

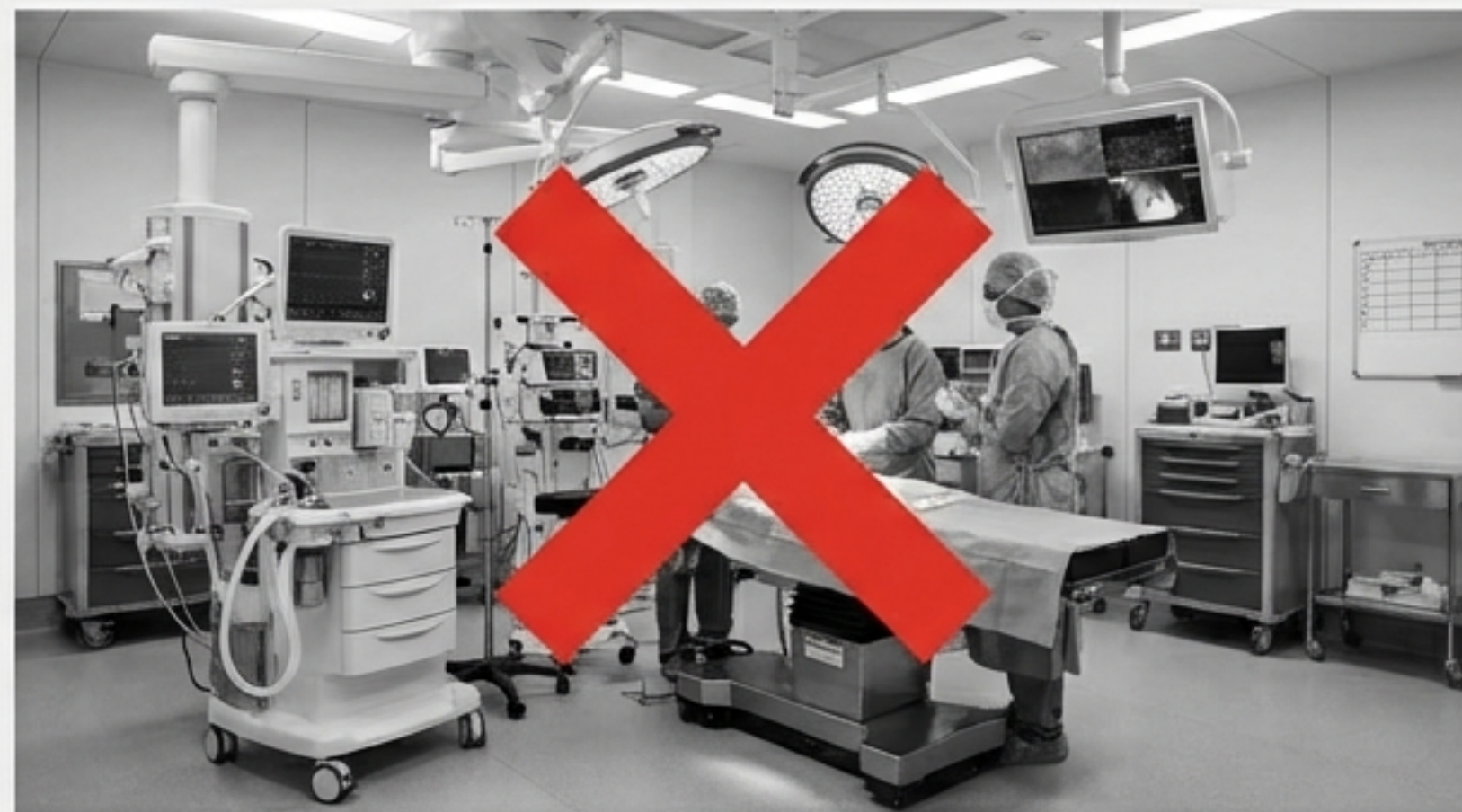
IGNORANCE

We fail because science has given us only a partial understanding of the world.



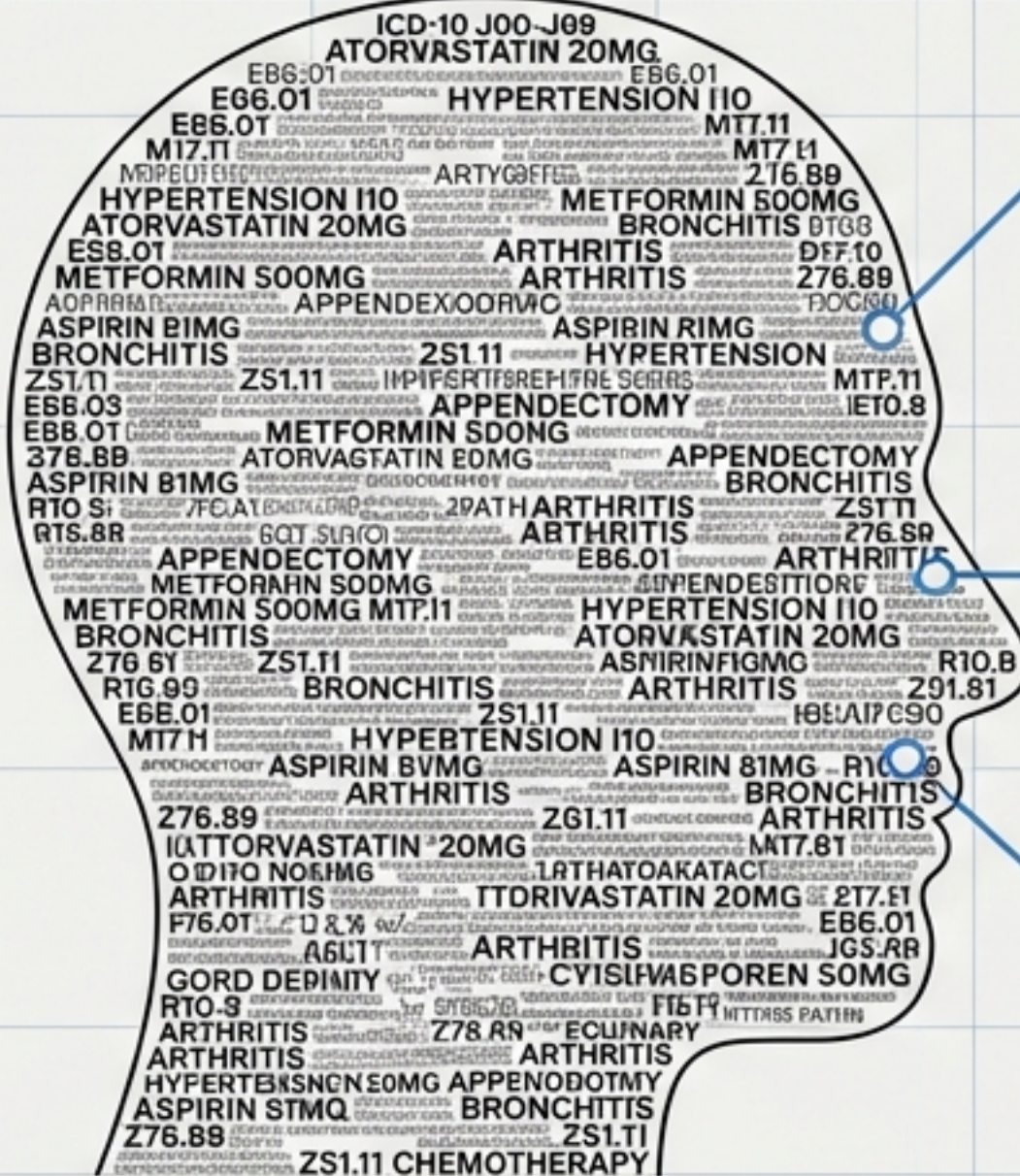
INEPTITUDE

The knowledge exists, but we fail to apply it correctly.



The 21st Century Shift: Our primary struggle is no longer Ignorance; it is Ineptitude.

The Burden of Knowledge



13,000 different diseases & injuries

6,000 drugs

4,000 medical & surgical procedures

The Kenmore Square Statistic

In a single year, general physicians evaluated an average of 250 different diseases and prescribed 300 different medications.

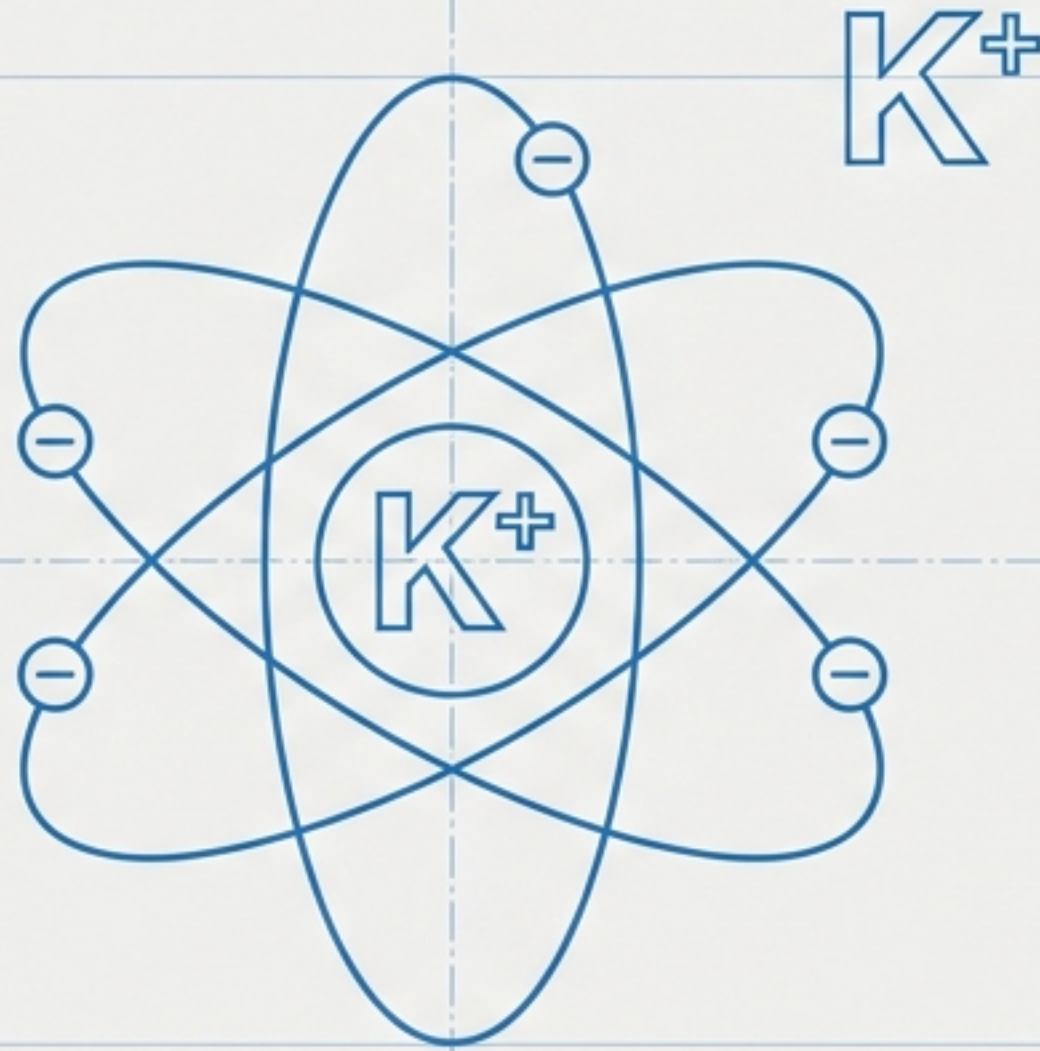
The volume of knowledge has exceeded the ability of any individual to manage it reliably from memory.



Expertise is Not a Safety Net

The Expert Team

- Surgeon
- Anesthesiologist
- Nurses
- Residents



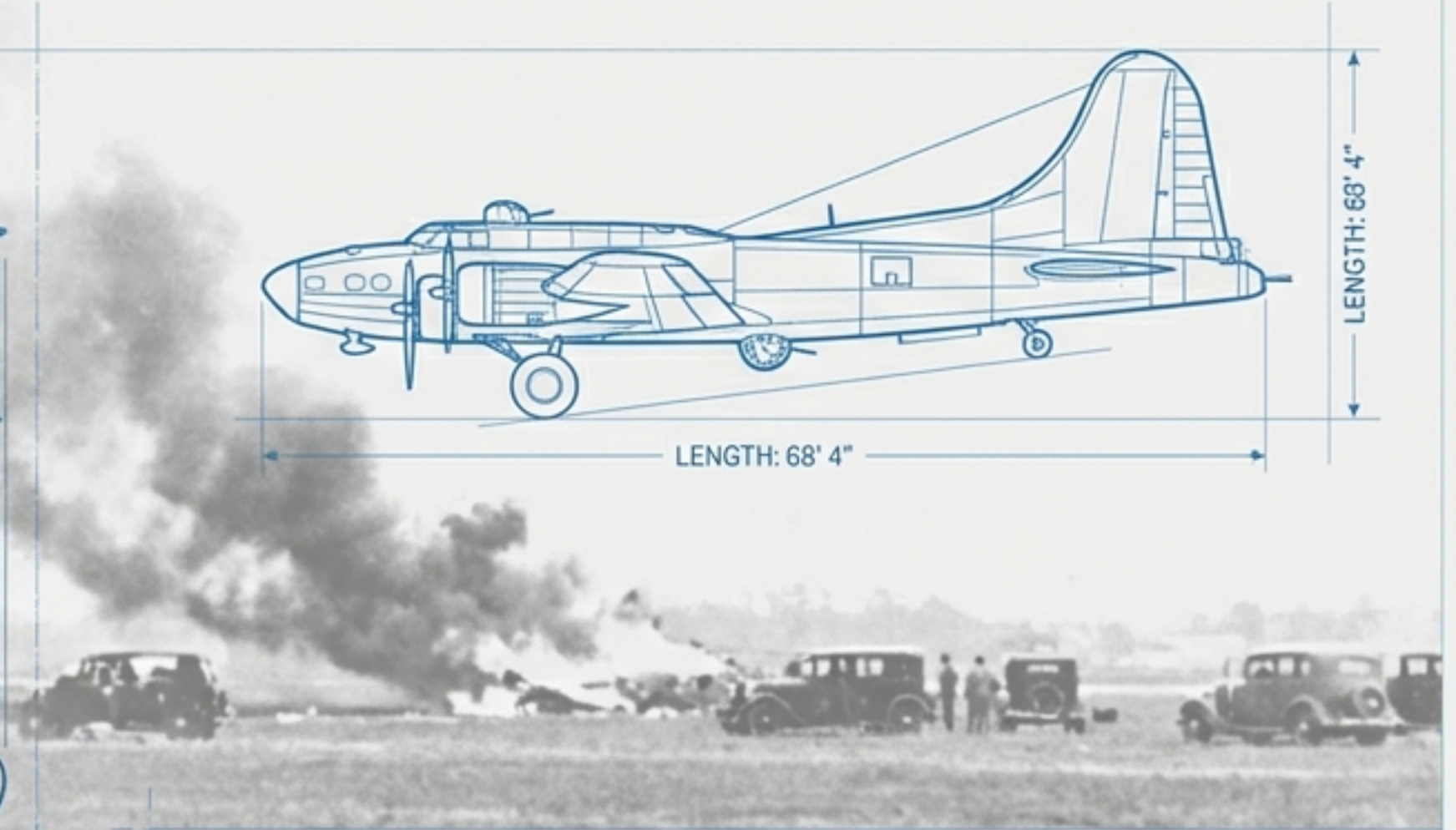
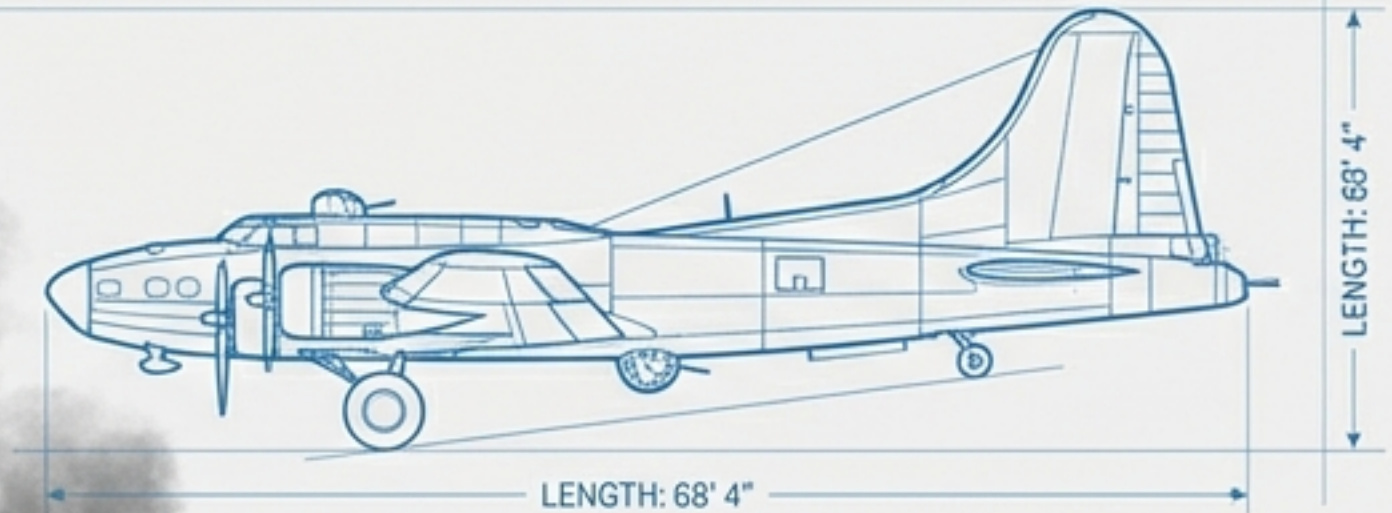
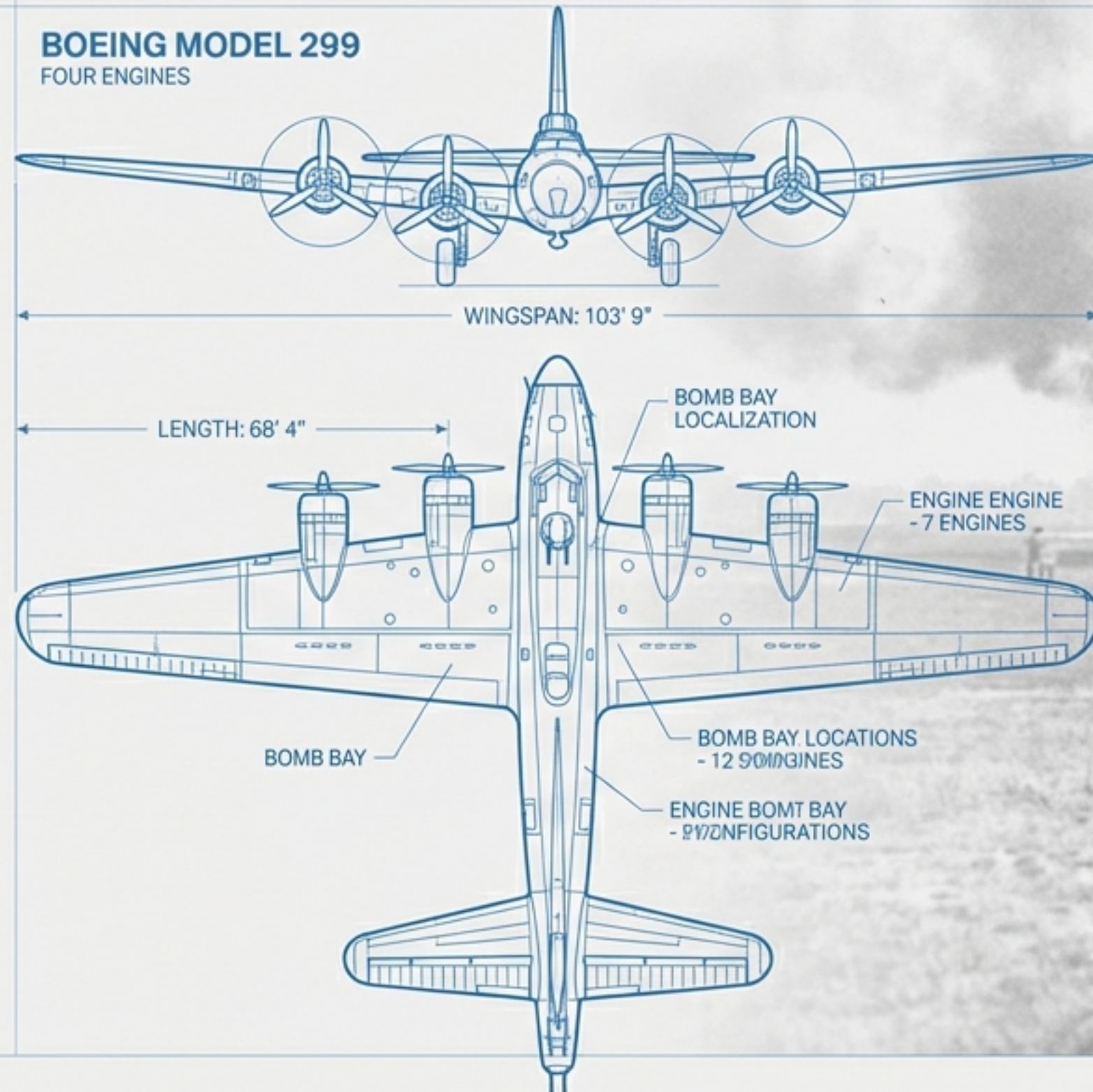
The Error

During a complex cancer surgery, the team missed a potassium overdose. A classic cause of cardiac arrest found in every textbook.

Under conditions of complexity, expertise is not enough.



October 30, 1935. Wright Air Field.



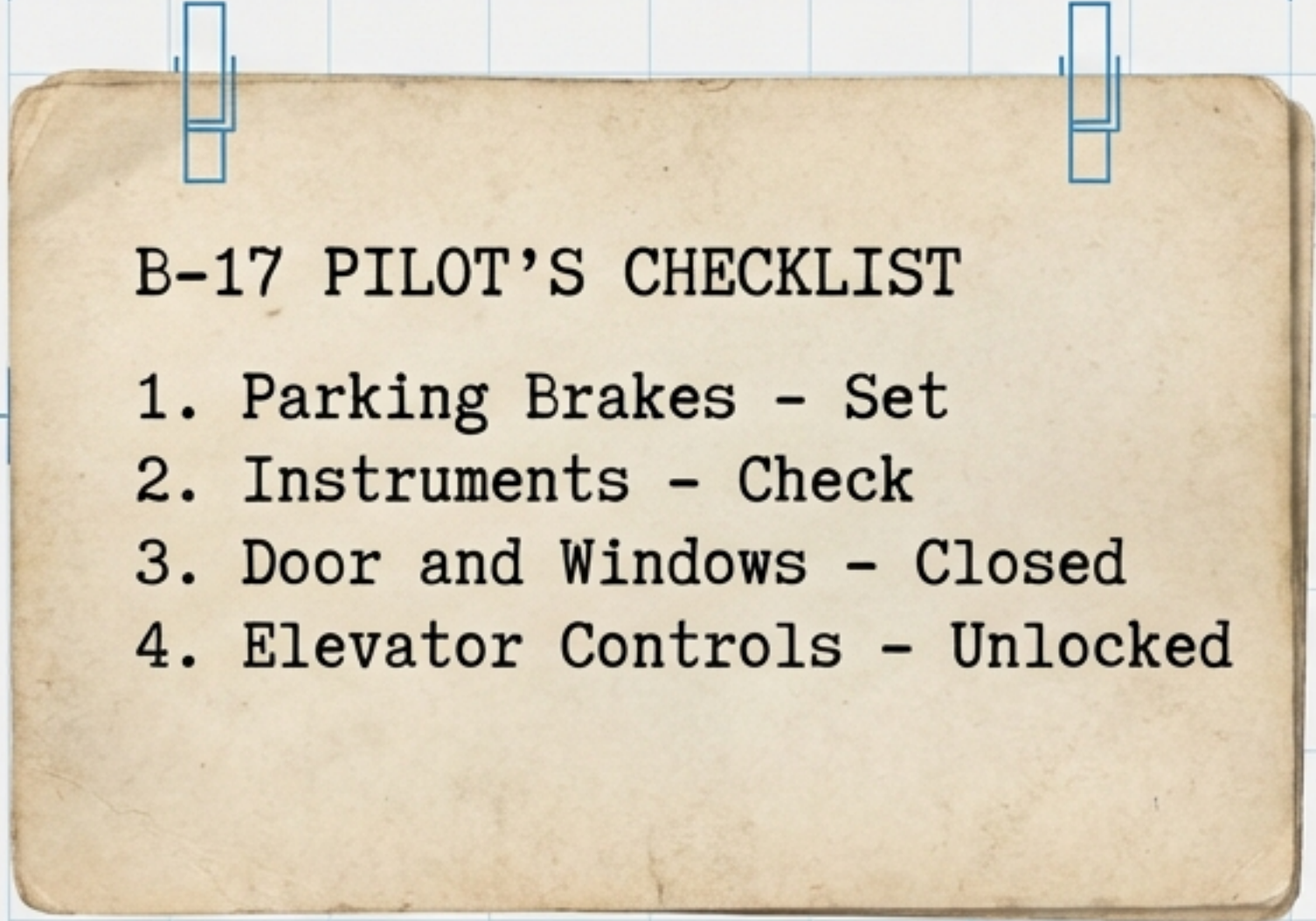
The Flying Fortress: Able to carry 5x the bombs and fly faster than any competitor.

The Failure: It stalled at 300 feet and crashed, killing Major Ployer P. Hill.

The Cause: Pilot Error. A forgotten elevator lock.



Too much airplane for one man to fly.



B-17 PILOT'S CHECKLIST

1. Parking Brakes - Set
2. Instruments - Check
3. Door and Windows - Closed
4. Elevator Controls - Unlocked

The Intervention: Not more training, but a checklist.

**Result: 1.8 million miles
accident-free.**

The Cognitive Net

The Discipline
of Higher
Performance:

Making the
minimum
necessary steps
explicit and
verifiable.

Memory
Loss

Distraction

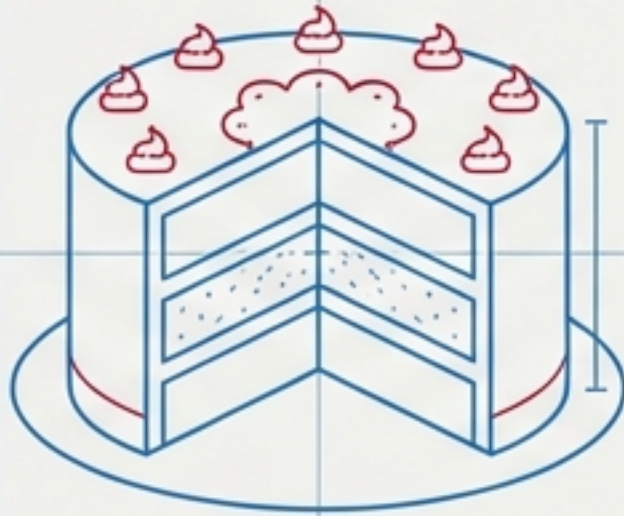
Skipping
Steps

Checklists catch mental flaws inherent in all of us.



Three Types of Problems

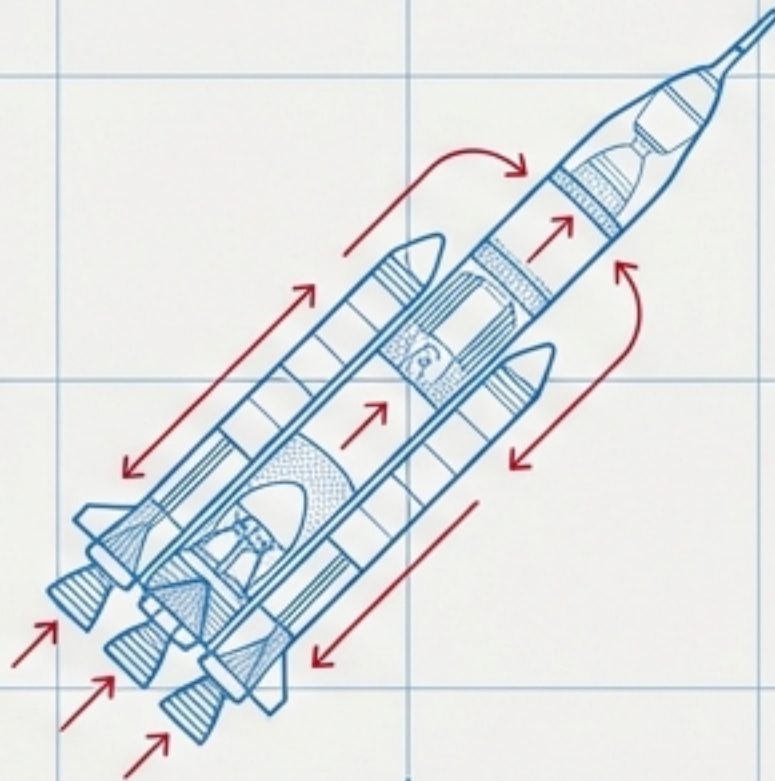
SIMPLE



Baking a Cake
Recipe-based.
Repeatable success.



COMPLICATED



Rocket to the Moon
Series of simple steps.
Requires coordination.



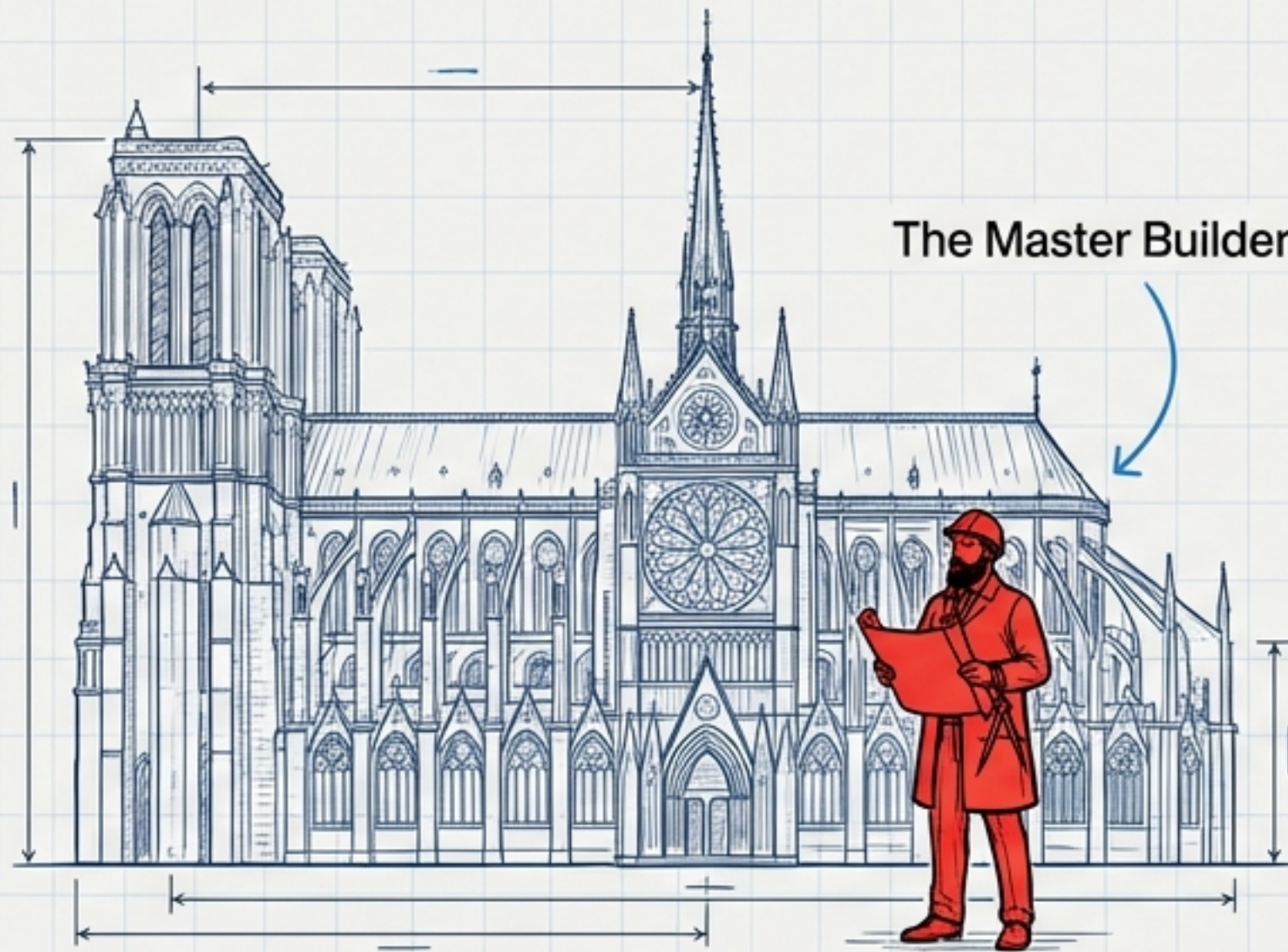
COMPLEX



Raising a Child
Unique. Outcome uncertain.
Formulas don't guarantee success.

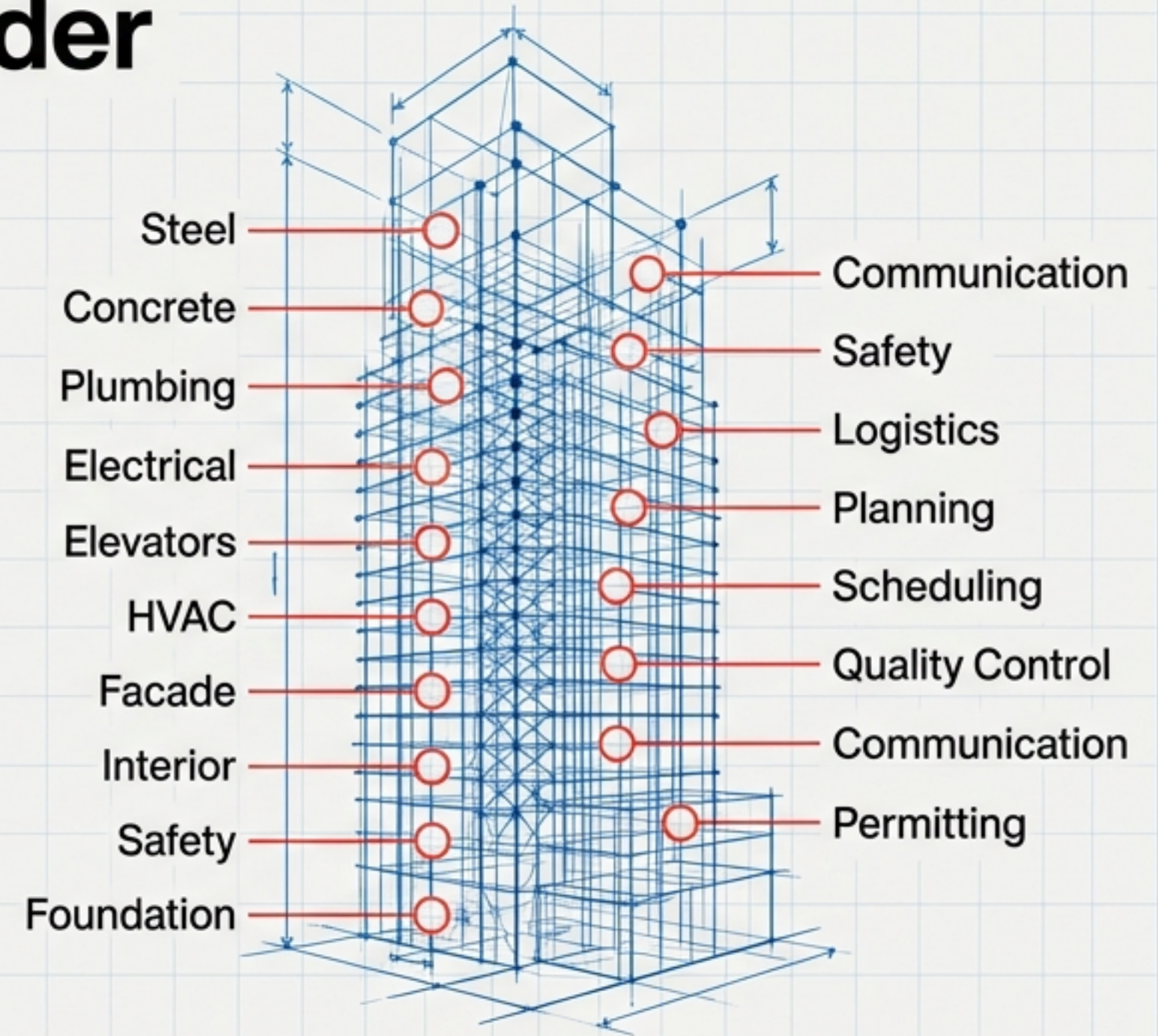


The Death of the Master Builder



The Master Builder

The Single Omniscient Leader



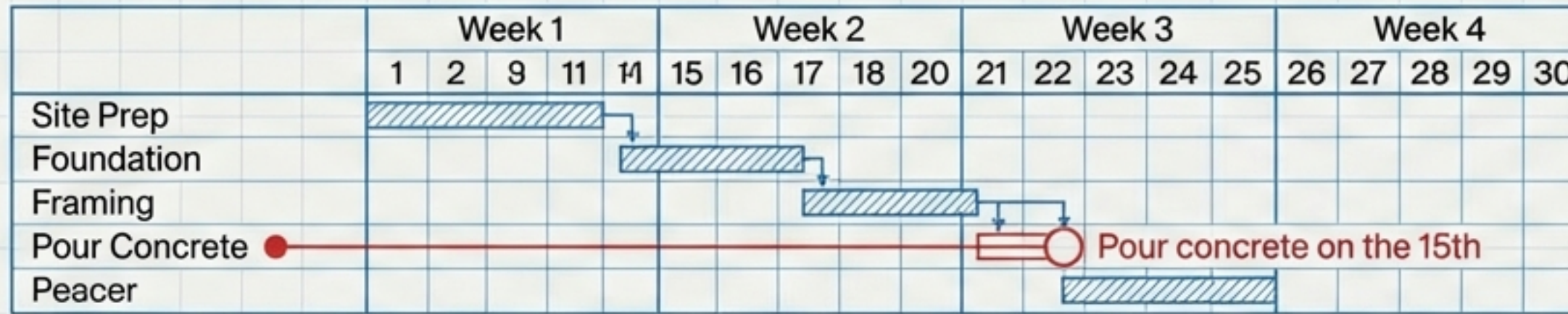
A System of Communication

We can no longer rely on one omniscient leader. Construction abandoned the Master Builder model for a system of communication.



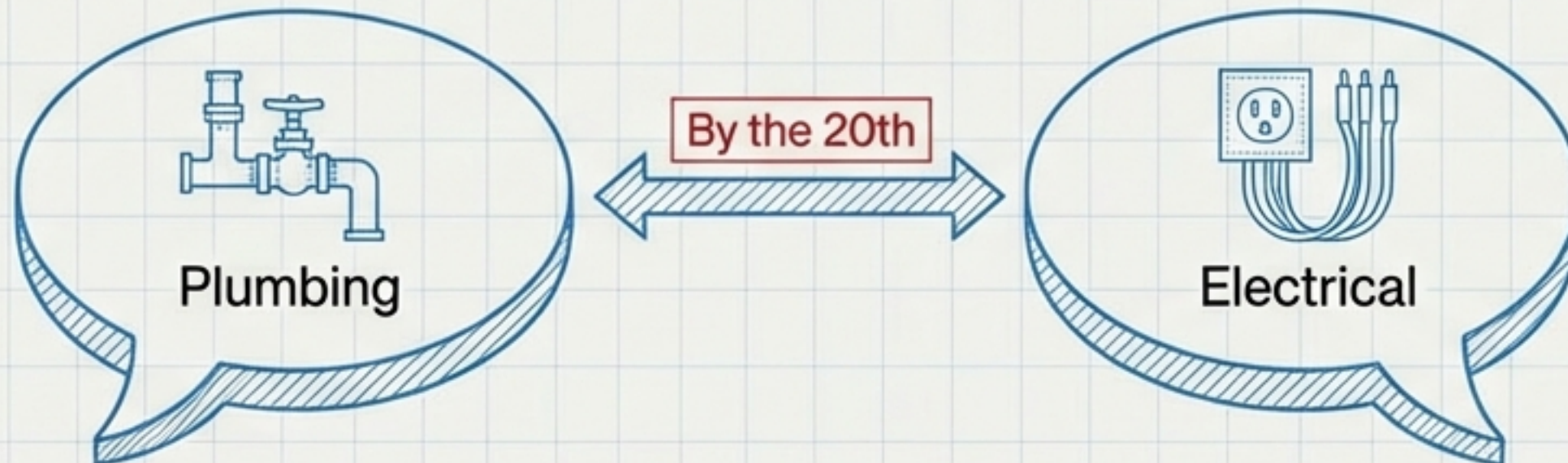
Two Types of Checklists

1. The Schedule (The “DO” List)



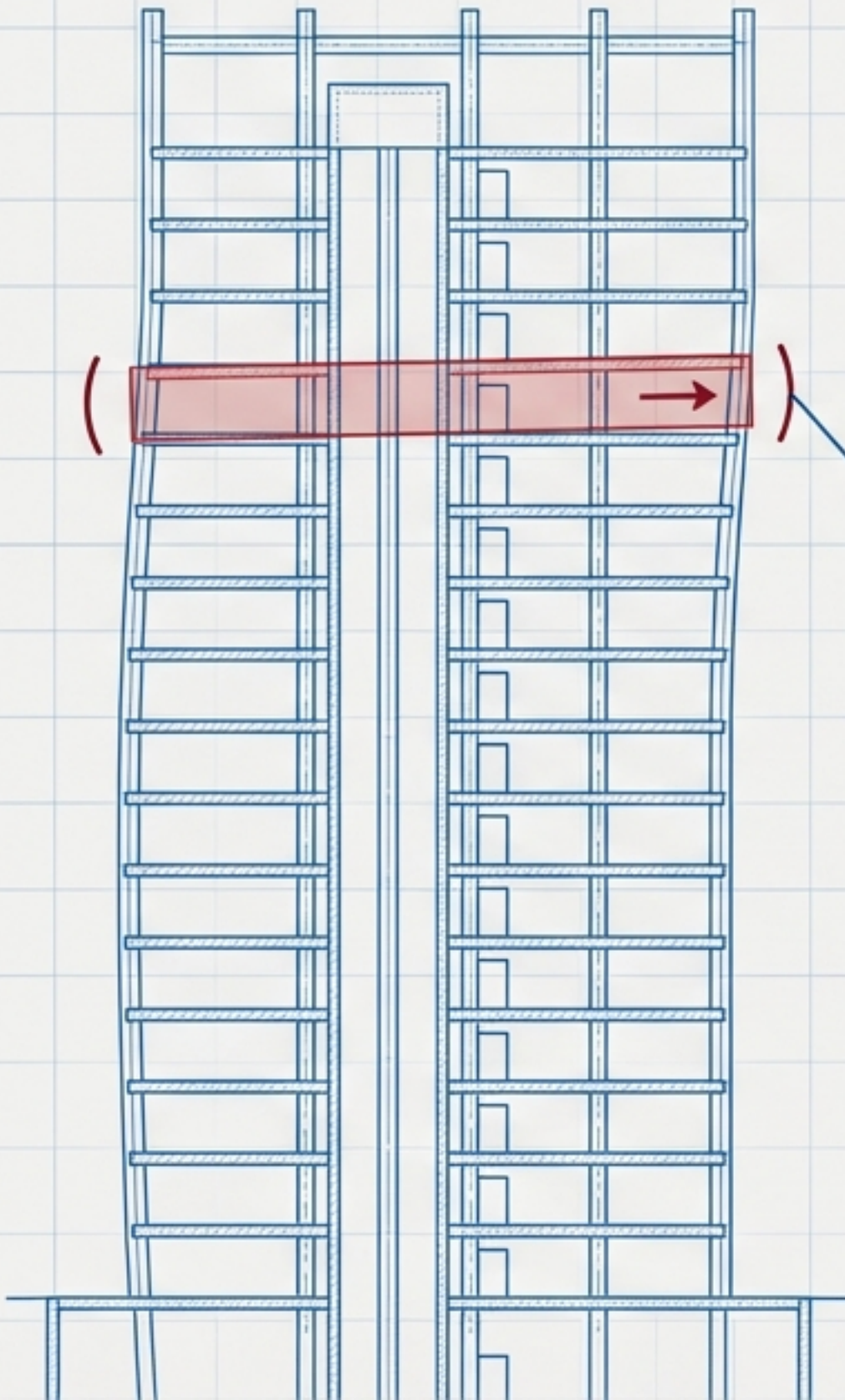
Forces simple tasks.
“Pour concrete on the 15th.”

2. The Submittal Schedule (The “TALK” List)

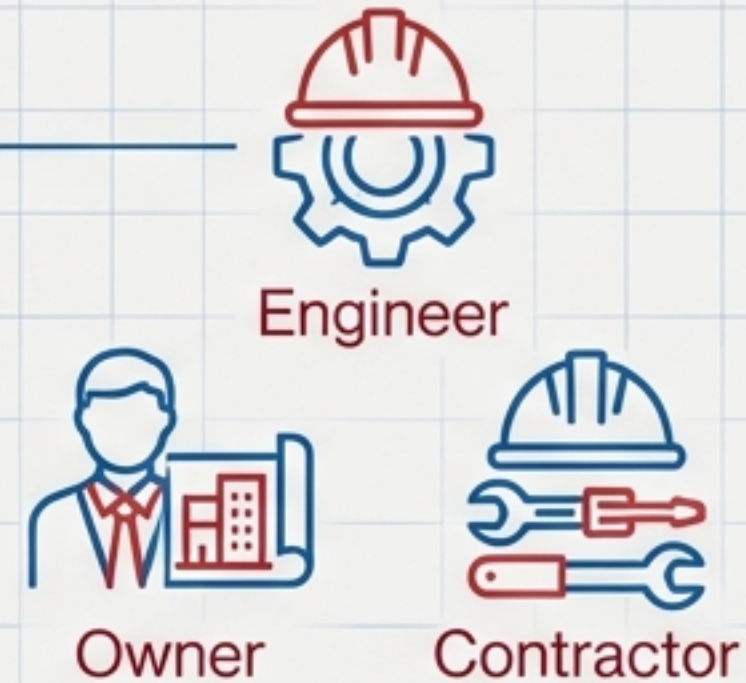


Forces communication.
“Plumbing must talk to Electrical by the 20th.”

Managing the Unexpected: The Tilting Floor



Problem: Settling caused upper floors to tilt.



Solution: The Submittal Schedule ensured they communicated. They adjusted the steel frame load.

The checklist didn't solve the problem; it ensured the experts talked so *they* could solve it.

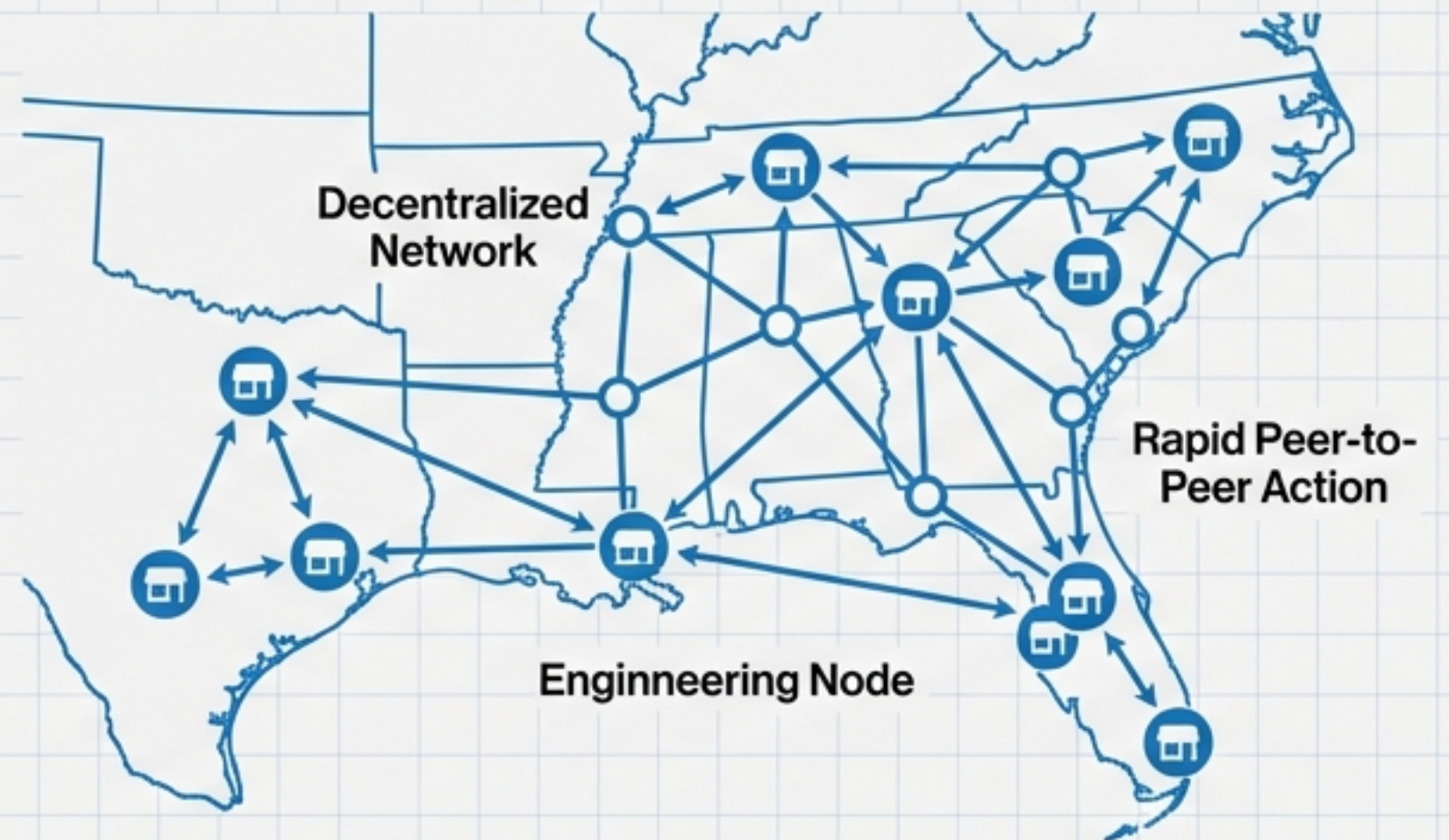


Pushing Power to the Periphery

Hurricane Katrina (Centralized Failure)



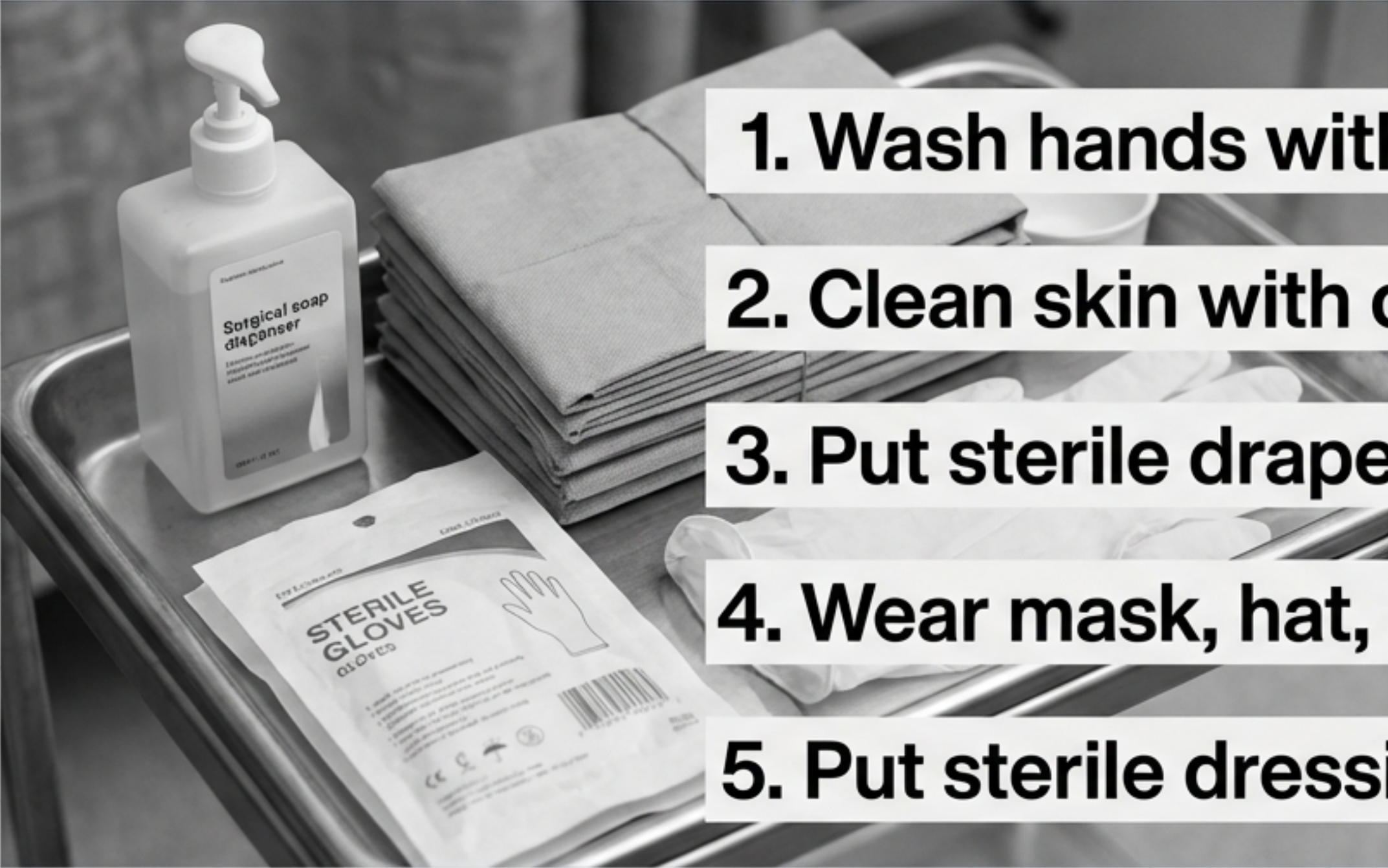
Wal-Mart Response (Decentralized Success)



"Make the best decision that you can with the information available to you." — Lee Scott, CEO



The Five-Step Intervention



1. Wash hands with soap.

2. Clean skin with chlorhexidine.

3. Put sterile drapes over entire patient.

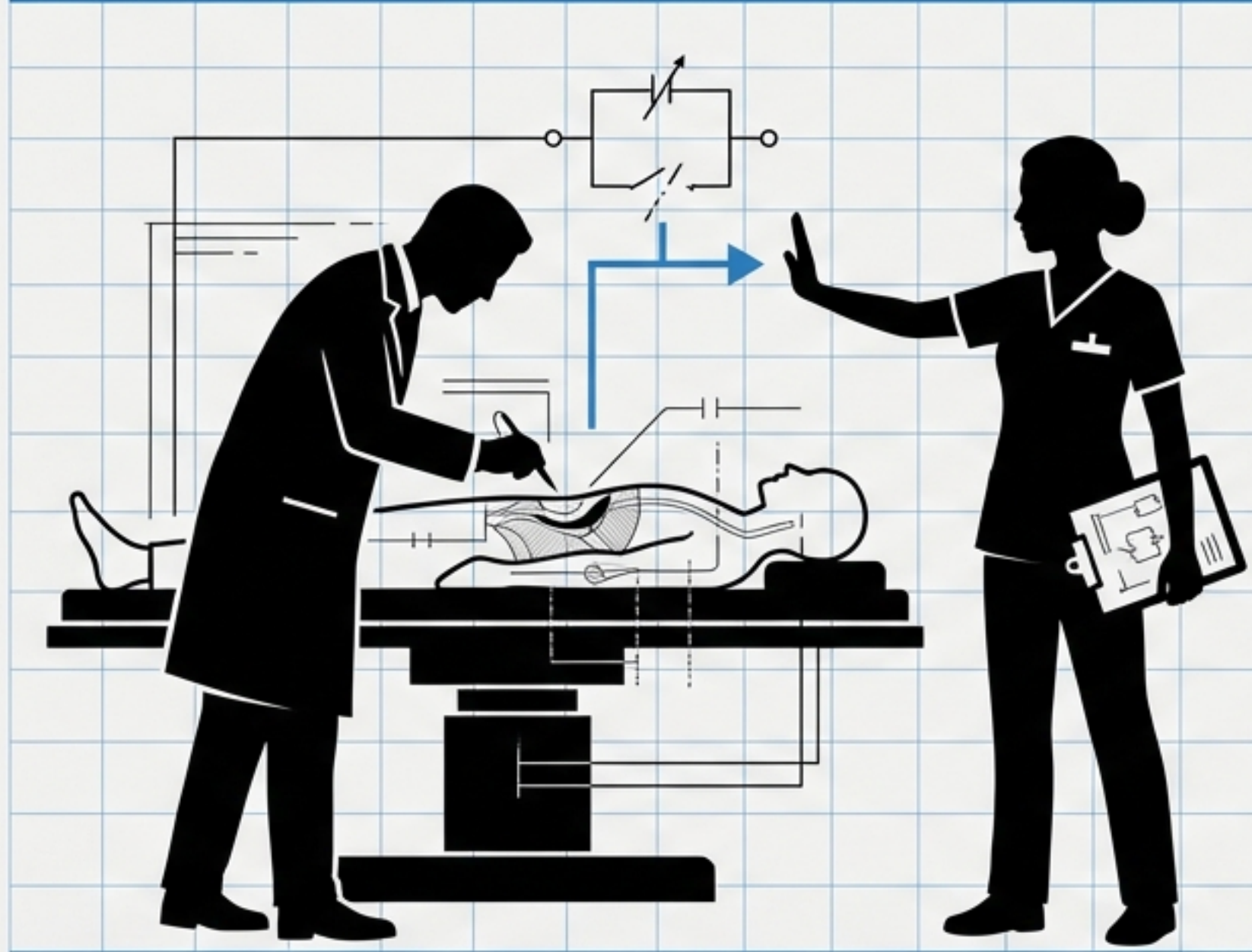
4. Wear mask, hat, sterile gown, gloves.

5. Put sterile dressing over insertion site.

In more than 33% of patients, experts skipped at least one step.



Changing the Power Structure



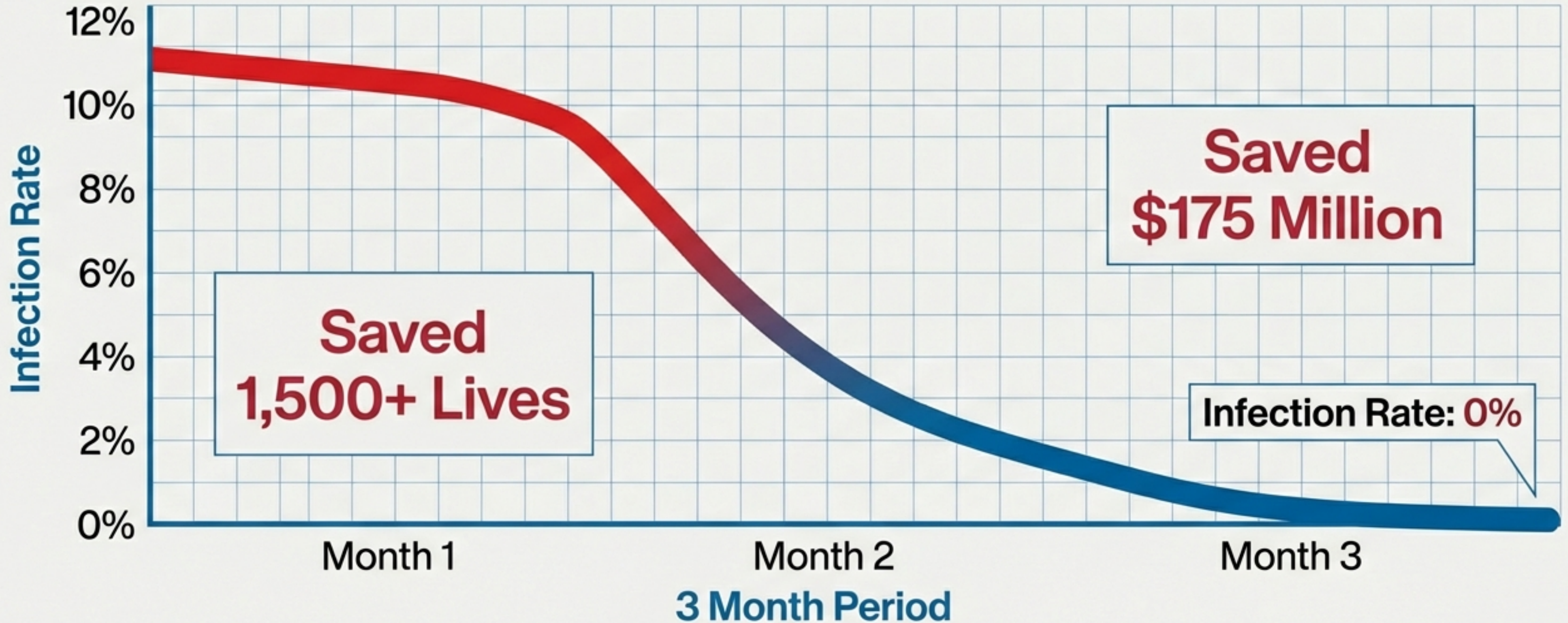
The Cultural Shift:

Administration
authorized nurses to
STOP doctors if a step
was missed.

From Autonomy (Master Physician) to Discipline (Teamwork).



The Keystone Initiative Results

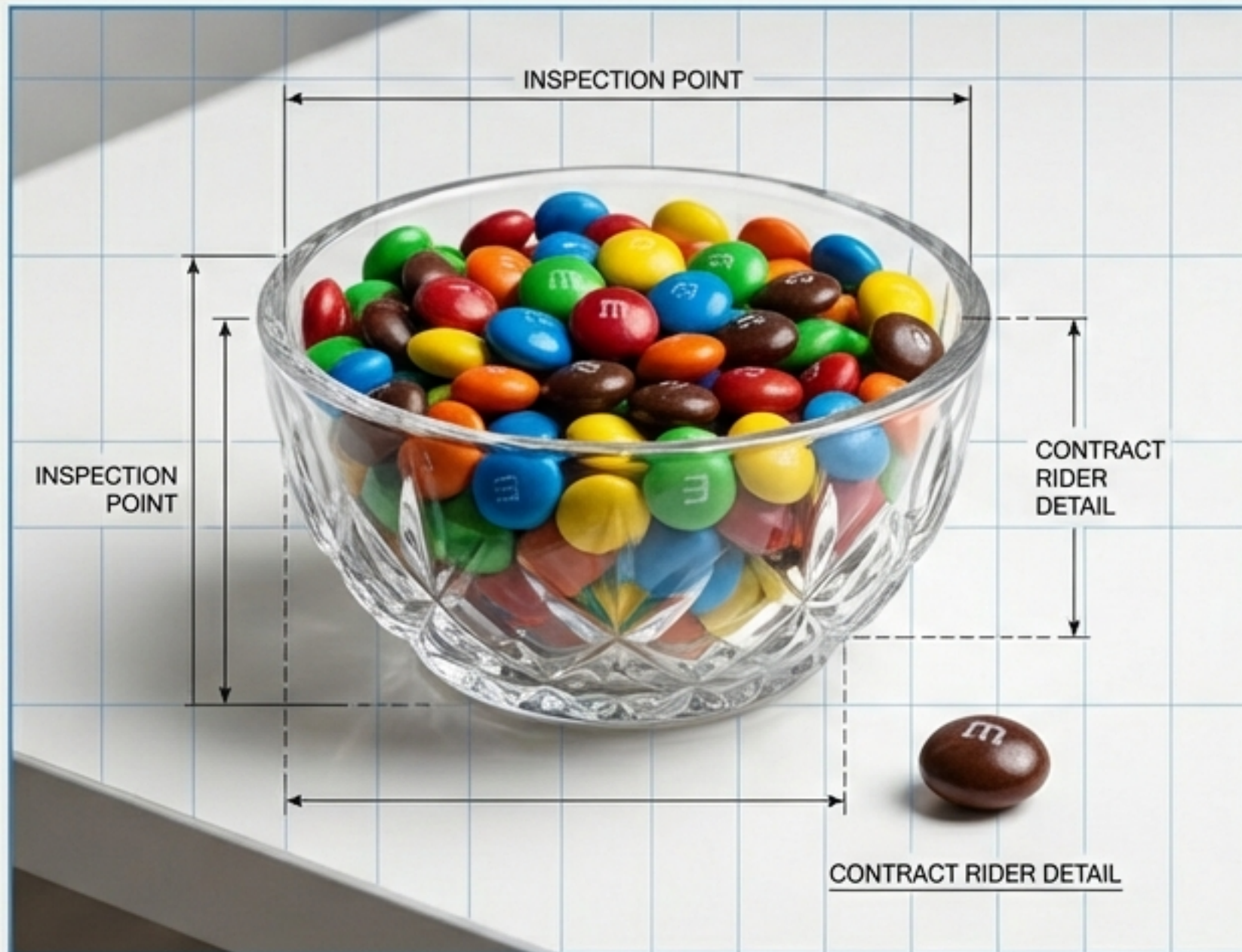


“All because of a stupid little checklist.”



The Brown M&Ms

Article 126: No Brown M&Ms.

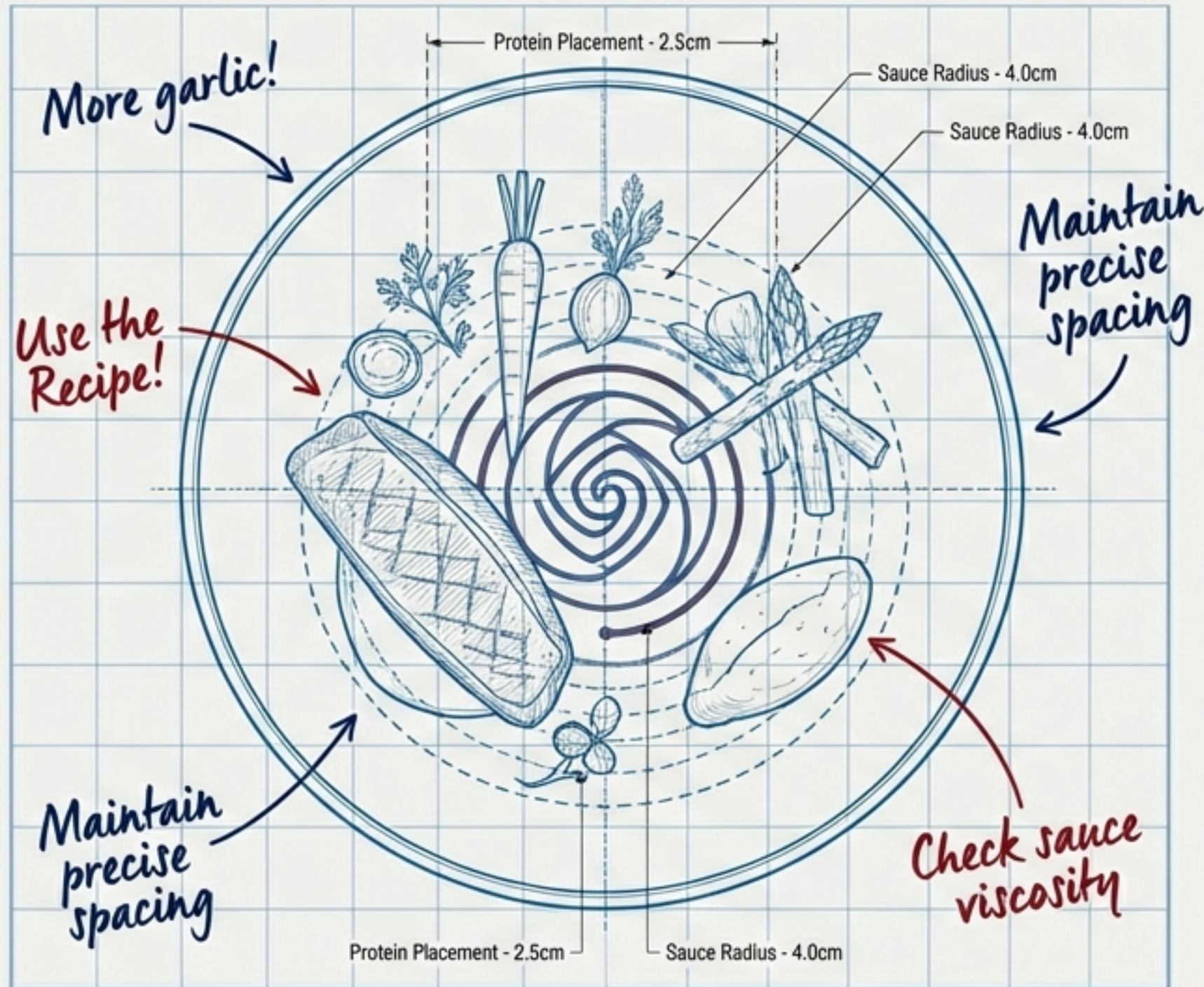


Van Halen's contract rider wasn't about being divas. It was a **tripwire**. If they saw a **brown M&M**, they knew the promoters hadn't read the safety specs.

A quick, **verifiable check** of a complex system's integrity.



Haute Cuisine & The Discipline of Craft



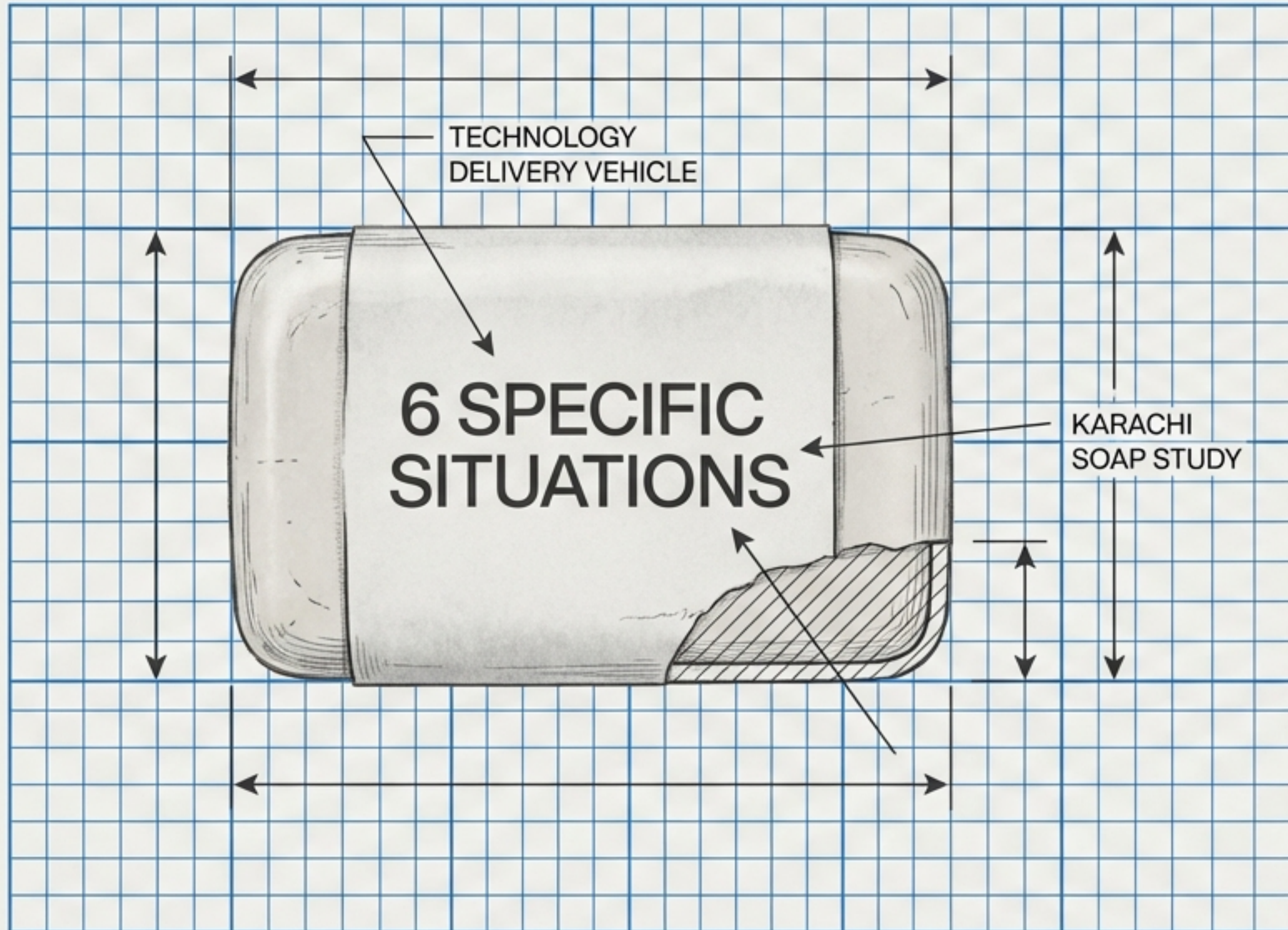
Rialto System:

1. The Recipe (The Do List)
2. The 5:00 PM 'Pow Wow' (The Talk List)
3. The Sous Chef Review (The Check)

Creativity is built on a foundation of consistency.



Simple. Measurable. Transmissible.



52% Drop
in Diarrhea

48% Drop in
Pneumonia

The Karachi Soap Study:
The soap was the technology.
The instructions were the
behavior-change delivery vehicle.



The Resistance



The Old Heroism (Autonomy).

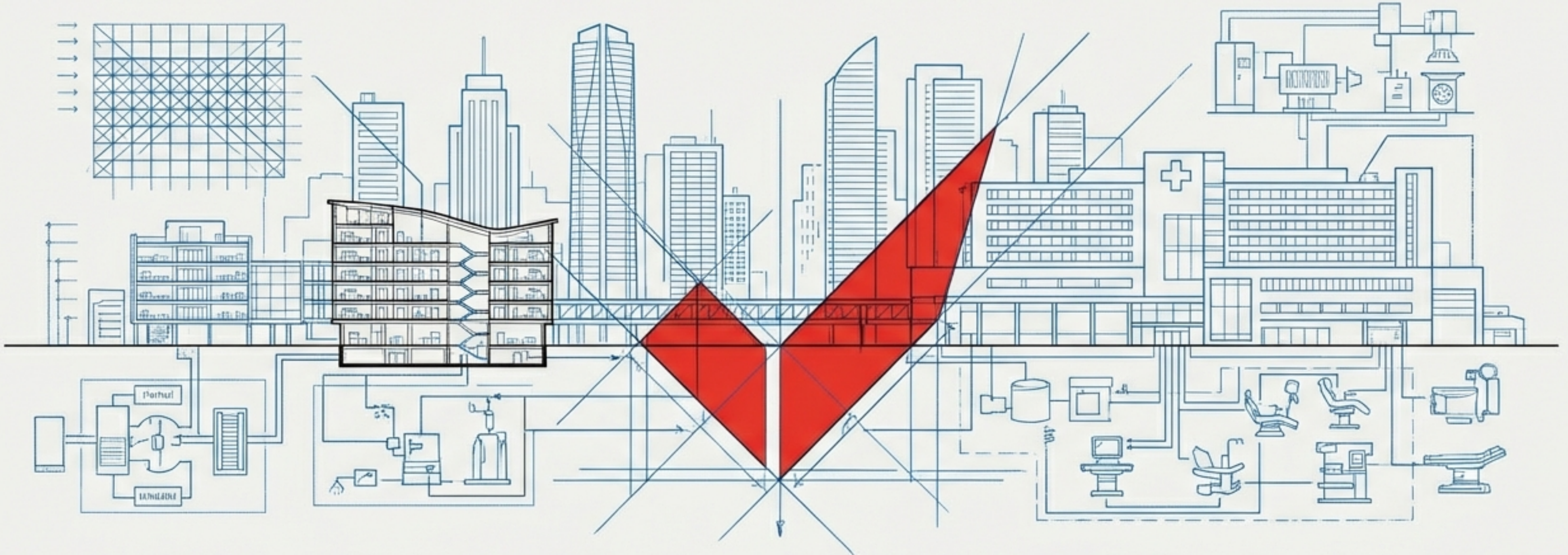


The New Heroism (Discipline).

We must stop preferring the 'Cowboy' to the 'Checklist'.
Even Sully used a checklist to land on the Hudson.



The New Ethic



- Acknowledge Fallibility
- Embrace Teamwork
- Utilize the Humble Tool

“Under conditions of complexity, not only are checklists a help, they are required for success.” ✓